Multi-Modal Person Re-ID & Face Association Pipeline

Real-Time Architecture for Retail Analytics

ENV	: Kaggle Edge
HARDWARE	: 2x NVIDIA Tesla T4 GPUs (16GB VRAM ea)
FRAMEWORK	: PyTorch / Python 3.12

MODALITIES : Body Re-ID

MODALITIES : Face Association

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System Mission & Core Operations

1. Detect



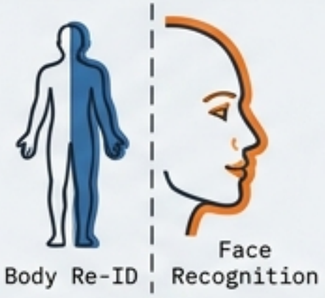
Scanning security footage to locate every human in every frame.

2. Track



Assigning a temporary `track_id` as targets move through the scene.

3. Identify



Matching visual appearances against a persistent database to assign a permanent `global_id`.

4. Record



Logging entry time, duration, and capturing highest-quality visual crops.

5. Search



Enabling sub-millisecond reverse-image lookups across millions of vectors.

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The Dual-Modality Paradigm

Body Re-ID

Strength:
Works at any distance/angle; does not require face visibility.

Weakness:
Fails when subjects wear similar clothing (e.g., store uniforms).

System Role:
Primary continuous identification.

Retroactive Ground Truth Override

Face Recognition

Strength:
Near-perfect accuracy; establishes definitive identity ground truth.

Weakness:
Demands close-up, front-facing, well-lit captures.

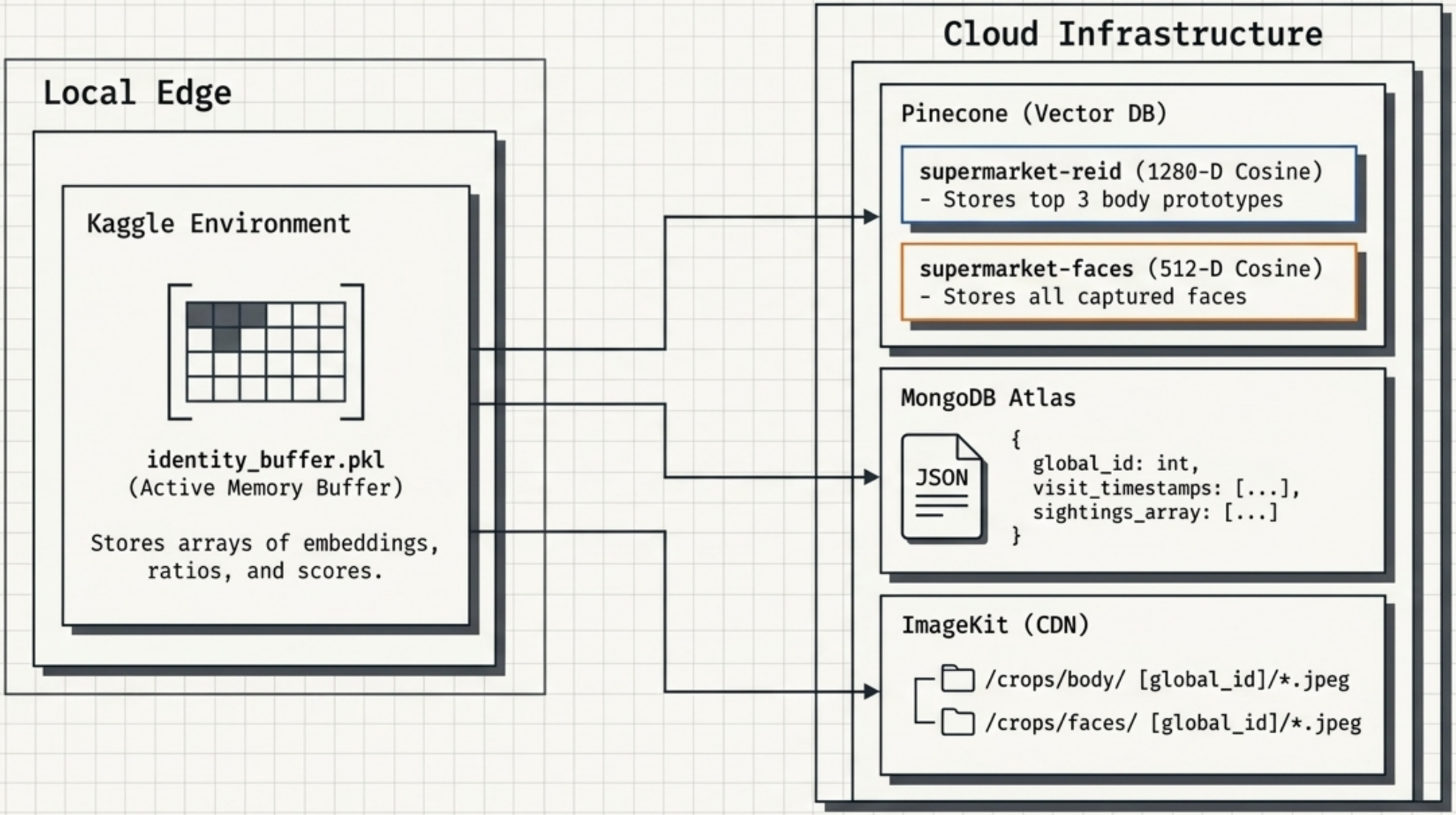
System Role:
Override & Correction.

AI Model Architecture & Specs

Model	Assignment	Specs	System Output
Detection & Tracking Pipeline			
RT-DETR-L	cuda:0	32-36M Params / 100-140 GFLOPs	**Person Detection.** NMS-free for predictable latency.
BotSort	cuda:0	OSNet_x0_25 ReID backbone	**Tracking.** Kalman filter state prediction.
SCRFD	cuda:0	640x640 Input	**Face Detection.** Outputs bounds & landmarks.
Heavy AI & Embedding Pipeline			
YOLOv11s-Pose	cuda:1	17 Keypoint Topology	**Quality Gate.** Validates skeleton visibility.
CLIP-ReID	cuda:1	ViT-B/16 / ~86M Params	**Body Embedding.** Outputs 1280-D L2-norm vector.
ArcFace	cuda:1	ResNet-50	**Face Recognition.** Outputs 512-D highly discriminative vector.

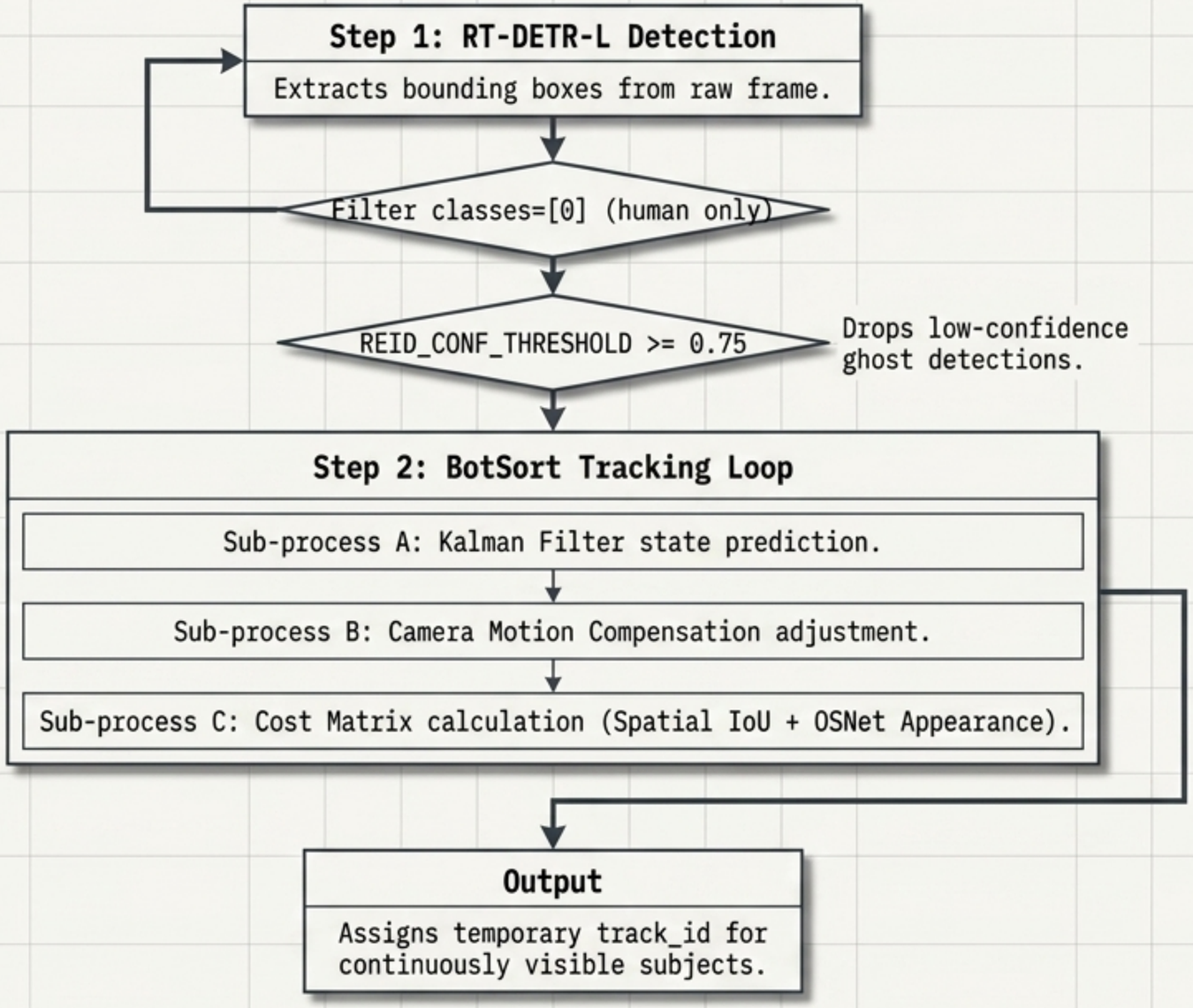
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Infrastructure & Data Flow



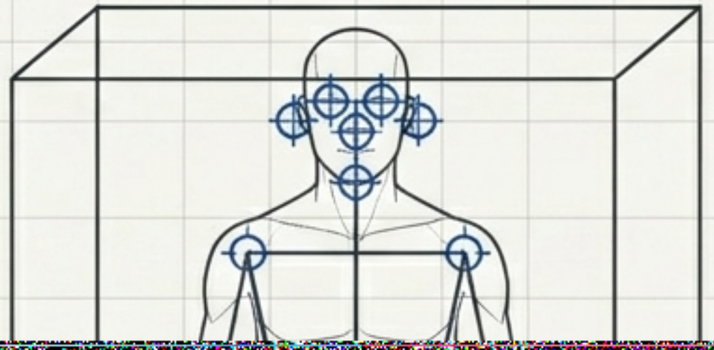
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The Frame Processing Loop (Phase 1)



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The Frame Processing Loop (Phase 2)

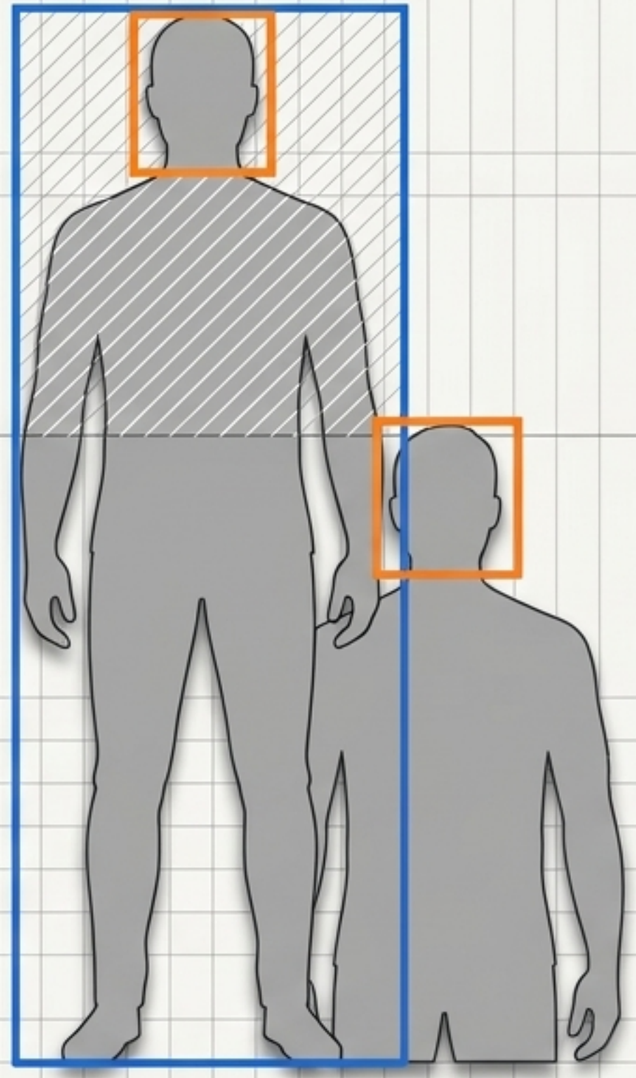


Step 4: The YOLOv11s-Pose Quality Gate

Calculate visible skeleton

Step 5: CLIP-ReID Feature Extraction

Spatial Pairing & Face Integration



The Problem

InsightFace detects faces independently. How do we accurately pair them to bodies in crowded frames?

The Top 40% Spatial Pairing Rule

An InsightFace detection must fall entirely within the upper 40% bounds of a tracked body bounding box to be successfully paired.

Prevents false pairings (e.g., matching a background face to a foreground torso).

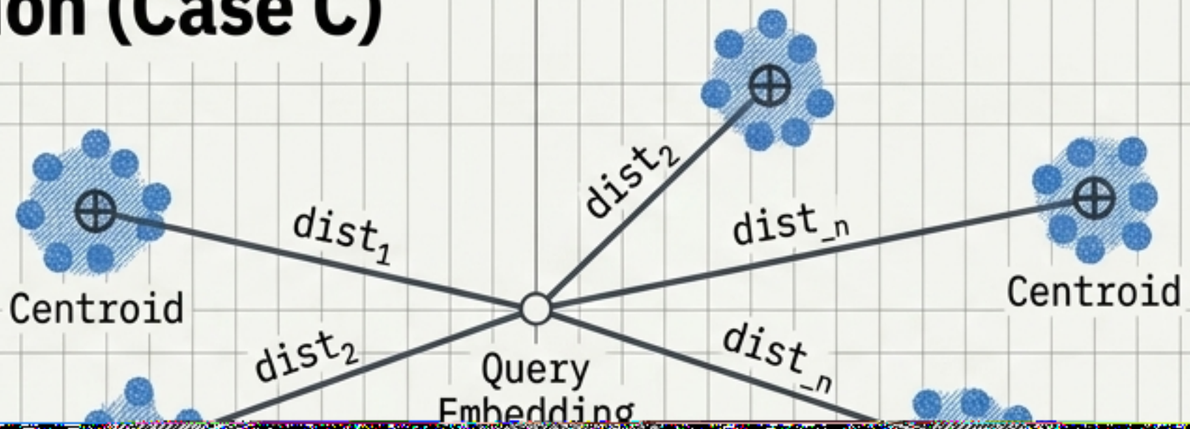
Face Pipeline Trigger

If successfully paired AND `face_det_score > 0.7`, extract `512-D ArcFace` embedding.

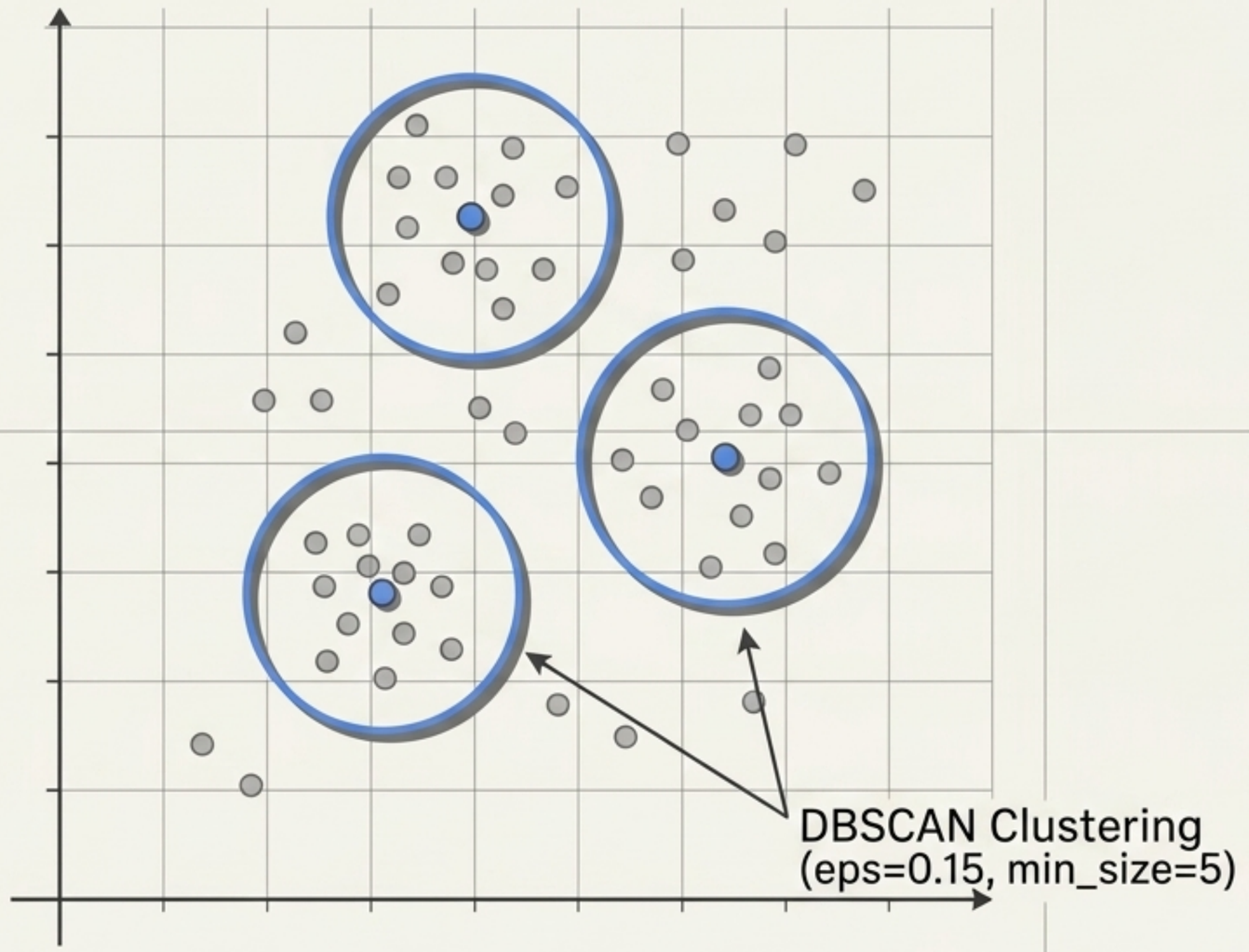
Query the `supermarket-faces Pinecone` index.

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Gallery Classification (Case C)



Managing Unknowns (UDM & Case D)



The UDM Trigger: Runs every 30 frames on crops that failed Cases A, B, and C.

Face VIP Bypass

If ANY crop in a cluster possesses a `forced_gid` from a face read, the entire cluster immediately inherits that identity via majority vote.

Merge Operation

If a newly formed cluster's centroid falls within mathematical `epsilon` of an existing gallery identity, merge them.

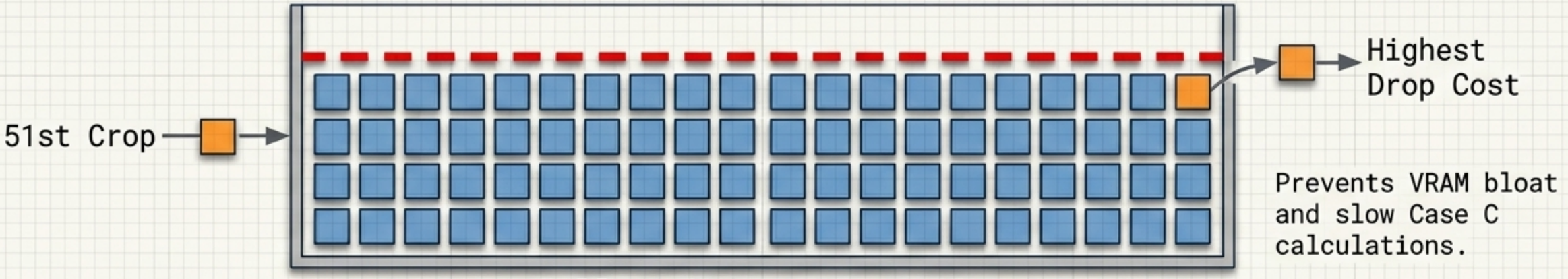
Mint Operation

If the cluster is entirely unique, mint a brand new `global_id`. Remaining noise points labeled `-1` stay in the pool.

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Self-Adaptive Gallery Budgets

MAX_IMAGES = 50 limit strictly enforced.



Drop Cost = Redundancy + Uncertainty

Redundancy
Max cosine similarity to any OTHER crop of the same person.
High score means it is visually identical to something we already have.

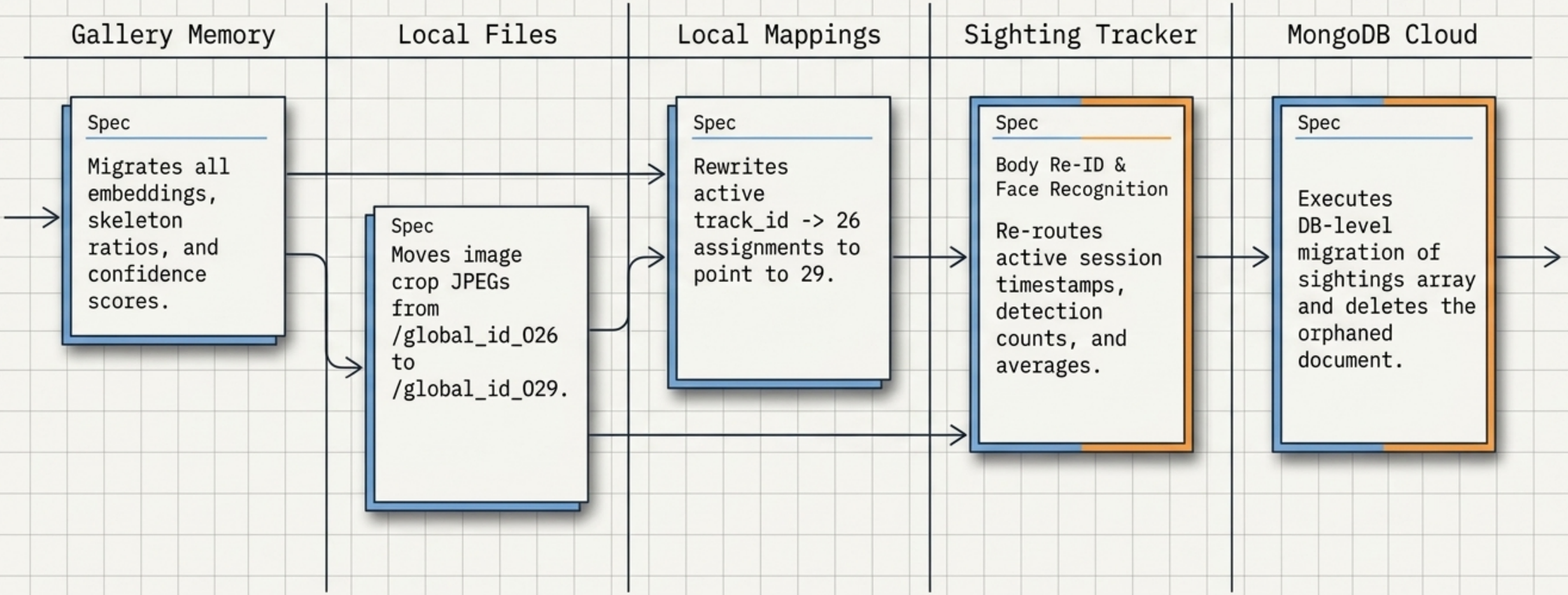
Uncertainty
Normalized entropy over softmax similarities to ALL gallery centroids.
High score means it is visually ambiguous and looks like multiple different people.

The system drops the worst crop: one that is both redundant AND confusing.

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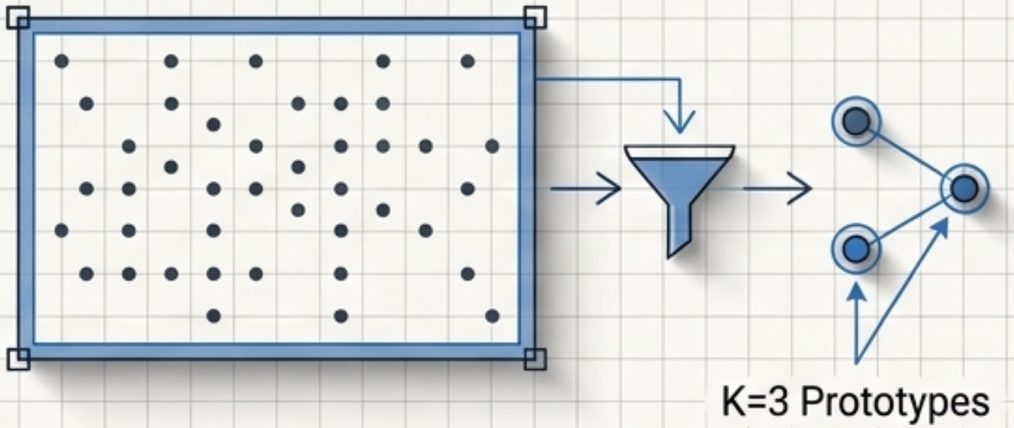
Identity Merge & Sighting Lifecycle

Trigger: execute_identity_merge (GID 26 merges into GID 29)



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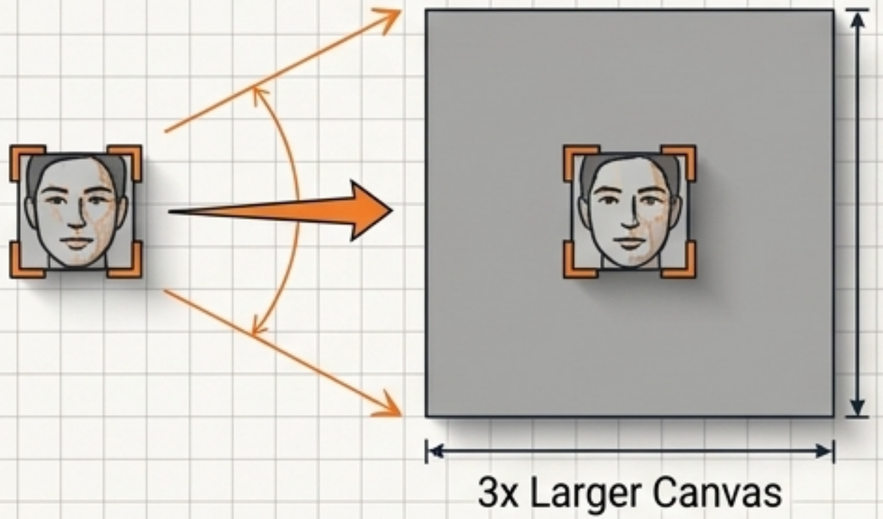
Search Engine Mechanics



Body Search Engine (Pinecone)

`Adaptive Farthest Point Sampling (a-FPS Prototypes)`

- Uploading all 50 crops per person is computationally wasteful.
- Algorithm selects K=3 prototypes per identity.
- Using `a = 0.5` balances quality (highest confidence) with diversity (visual variance: front/side/back views).



Face Search Engine (Pinecone)

`Direct ArcFace 512-D Query`

- ArcFace 512-D vectors are highly discriminative. Direct lookup, no prototypes needed.
- **Edge Case Handle:** Pre-cropped face chips uploaded by users fail InsightFace detection (no background context).
- System automatically pads the crop into a 3x larger gray canvas to force detection.

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System Evaluation & Engineering Constraints

Performance Readout (Brute Force)

85.35% Accuracy

1,124 / 1,317 accurate reverse searches

Engineering Constraints & Realities

RAM-First Processing

Optimized for Kaggle constraints. Sightings pool in memory and bulk-upload post-video. Vulnerable to mid-processing kernel crashes.

State Consistency

